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Introduction

1. Rationale

The accelerating shift toward hybrid work arrangements, coupled with breakthroughs in artificial intelligence, has profoundly transformed how organizations manage their human resources. What began as an IT enhancement has evolved into a foundational business strategy. According to McKinsey & Company (2025), firms that integrate digital HR practices consistently deliver productivity improvements far exceeding those of companies clinging to conventional methods. This compelling evidence explains why the vast majority of contemporary enterprises have transitioned their essential HR operations to scalable cloud-based platforms. However, acquiring advanced software alone falls short of delivering transformative results. Gartner's analysis (2024) reveals that only a small percentage of HR departments extract the full business value from these investments. The root cause lies in a critical oversight: true digital evolution necessitates a complete overhaul of workflows, rather than merely automating persistent manual inefficiencies.

These worldwide dynamics are vividly evident in Vietnam's evolving business landscape. The government's ambitious National Digital Transformation Programme has ignited a nationwide drive to modernize corporate operations and nurture a workforce proficient in digital tools (OpenGov Asia, 2021). For Vietnamese companies, this initiative imposes urgent pressure to revolutionize human capital strategies. As detailed in the Vietnam Trade and Industry Review, HR digitalization here has progressed from rudimentary administrative tasks to sophisticated data analytics and seamless integration with broader business objectives (Phan, T. C. M. 2024). Nevertheless, numerous domestic enterprises linger in a state of incomplete transition. They contend with outdated legacy systems and a patchwork of disparate software solutions, which spawn isolated data silos and obstruct a unified perspective on organizational performance.

HPA Direct VN Company Limited stands as a quintessential case study of these pervasive operational hurdles. Currently, the company handles vital HR functions, from employee onboarding and timekeeping to intricate payroll computations, through fragmented manual procedures and isolated spreadsheets. Such siloed operations inevitably produce severe inefficiencies. They consume excessive administrative effort, heighten the risk of human errors, and compromise data transparency across the organization. In the absence of a centralized database or real-time analytics tools, the Board of Directors operates without essential visibility into key performance indicators, including overtime expenditures and employee turnover trends. For HPA Direct VN, the core challenge extends beyond selecting a

single software tool. Instead, it demands a holistic integration of technology into everyday processes, ensuring an uninterrupted, automated data pipeline that spans from initial employee records to final payroll disbursements.

Overcoming these entrenched issues calls for a strategic, all-encompassing solution. Deploying a customized Enterprise Resource Planning (ERP) framework emerges as an imperative step, aligning the company with global benchmarks and its own expansion ambitions. This research, titled "Digital Transformation of HR and Payroll Processes using ERP: Current State Analysis and Solution Proposal for HPA DIRECT VN COMPANY LIMITED" directly addresses this need. It conducts a rigorous examination of the company's present administrative constraints, reimagines its primary HR workflows for the digital era, and recommends a cohesive ERP platform augmented by an interactive Business Intelligence dashboard. Ultimately, this approach promises to streamline resource allocation, safeguard data accuracy and compliance, and equip leadership with real-time, actionable insights. These enhancements will empower HPA Direct VN to cultivate a more agile, competitive, and future-proof organization.

2. Research objectives

2.1. General Objective

The overarching objective of this thesis is to comprehensively evaluate the current manual Human Resources and Payroll data flows at HPA DIRECT VN COMPANY LIMITED. Consequently, the research aims to propose a solution that restructures business processes and designs an integrated ERP system architecture coupled with an automated management dashboard. This solution strives to eliminate operational bottlenecks caused by manual execution, comprehensively digitize HR data, and provide data visualization tools to empower the Board of Directors in making data-driven strategic decisions.

2.2. Specific Objectives

To successfully achieve the overarching goal, this research strictly focuses on three interconnected specific objectives. First, the study aims to optimize and reengineer the existing business processes by investigating the limitations of the current (AS-IS) workflows and applying the Business Process Reengineering (BPR) methodology to design streamlined target (TO-BE) workflows, thereby ensuring seamless data integration across the HR Records, Time & Attendance, and Payroll sub-modules. Second, the research seeks to specify and design the ERP system architecture by analyzing functional requirements and utilizing standard international modeling notations, specifically UML and BPMN 2.0, to construct system Use Cases

and establish a robust Role-Based Access Control (RBAC) matrix for end-users. Finally, the thesis endeavors to develop an automated management dashboard by identifying critical Human Resource Key Performance Indicators (HR KPIs), enabling real-time data extraction and visualization to actively support the enterprise's cost control and strategic human resource administration.

3. Research Questions

To systematically achieve the established research objectives and construct a holistic digital transformation solution, this study is fundamentally guided by three core analytical questions. First, what are the critical operational bottlenecks and inherent risks within the current manual human resource and payroll workflows at HPA DIRECT VN COMPANY LIMITED, and how can these processes be structurally optimized using the Business Process Reengineering methodology to achieve seamless cross-departmental data integration? Second, addressing the identified inefficiencies, what are the precise functional requirements, and how should the architecture of the proposed Enterprise Resource Planning solution be mathematically modeled to guarantee secure, centralized data management alongside a strict Role-Based Access Control matrix? Finally, to empower data-driven executive leadership, which specific Key Performance Indicators are most vital for tracking organizational efficiency, and how can these workforce metrics be dynamically visualized through an automated management dashboard?

4. Object and Scope of the Study

4.1. Object of the Study

The primary object of this research is the digital transformation of Human Resources and Payroll workflows through the application of an Enterprise Resource Planning (ERP) system. Specifically, the study analyzes the underlying business processes, data flows, and the subsequent architectural design applied to the human resource management activities at HPA LLC Company from 2024 to the present.

4.2. Scope of the Study

The theoretical scope of this thesis covers the fundamental principles of management information systems and ERP frameworks. It specifically applies the methodologies of Business Process Reengineering (BPR) along with system analysis and design based on the object oriented approach. The research also utilizes standard international notations, including the UML and BPMN, to visually construct the administrative workflows.

In terms of practical application, the study examines the history of establishment, organizational structure, and core business operations of HPA DIRECT VN COMPANY LIMITED. It focuses on assessing the current state of human resource administration and payroll practices at the company from 2024 to the present. Driven by these empirical observations, the research details the necessary steps to analyze and design a comprehensive software solution. This specific design encompasses essential modules for personnel records, time attendance, payroll calculation, and a management dashboard tailored to the operational needs of the enterprise.

5. Research methodology

To systematically achieve the research objectives, this study employs a combination of qualitative data collection methods to ensure a deep understanding of the organizational context. The primary data is gathered through direct expert interviews with key stakeholders at HPA DIRECT VN COMPANY LIMITED. Specifically, detailed discussions are conducted with the Human Resources Manager and relevant administrative staff to uncover the practical challenges, hidden operational bottlenecks, and specific expectations regarding the new digital system. In addition to these interviews, the research utilizes document analysis to collect vital secondary data. This involves a thorough review of the existing internal policy manuals, daily attendance records, and legacy spreadsheet templates currently used for payroll calculation. By analyzing these internal documents, the study establishes a solid empirical foundation to accurately evaluate the current state of human capital management.

Following the comprehensive data collection phase, the research applies specific Business Analysis methodologies to bridge the gap between the current operational state and the target system architecture. The study utilizes the BPMN standard to visually map the existing workflows and pinpoint areas of severe inefficiency. This specific modeling technique is then applied to reconstruct and design the optimized future processes, ensuring a logical flow of information across all administrative departments. Furthermore, the research conducts a rigorous system requirements analysis to translate the identified business needs into concrete technical specifications. This technical phase involves defining both functional and non functional requirements, followed by modeling the structural framework of the ERP solution using the UML. By integrating precise process modeling with professional business analysis techniques, the study ensures that the proposed software architecture strictly aligns with the strategic objectives of the enterprise.

6. Structure of the Study

The thesis is organized into three main sections: the Introduction, the Main Content, and the Conclusion. The core content of the research is divided into three specific chapters:

Chapter 1: Theoretical Framework of Human Resource Management Information Systems.This chapter focuses on the fundamental theories and concepts related to Human Resource Management Information Systems (HRIS). It establishes the necessary academic background and explores the roles and components of these systems in modern organizational management.

Chapter 2: Analysis and Evaluation of Human Resource Management Analysis and Evaluation of Human Resource Management at HPA Direct VN LLC. This chapter provides an overview of the case study company and conducts a detailed analysis of its current human resource management practices. By evaluating the existing processes, the chapter identifies strengths, weaknesses, and key issues that need to be addressed through digital transformation.

Chapter 3: Development Orientations and Proposed Solutions for Human Resource Management at HPA Direct VN LLC. Based on the findings from the previous chapters, this chapter outlines the development strategies and proposes practical solutions to improve the human resource management system. It focuses on recommending specific ERP-based or digital tools tailored to the company's needs to enhance operational efficiency.

CHAPTER 1: THEORETICAL FRAMEWORK OF HUMAN RESOURCE MANAGEMENT AND ERP SYSTEMS

The landscape of human resource administration is currently undergoing a significant operational transition. For decades, traditional personnel management was viewed primarily as an administrative cost center characterized by heavy transaction processing, rigid organizational hierarchies, and paper based compliance tracking. Today, digital transformation has fundamentally redefined this discipline. Digital human resource management is defined as the deep assimilation of advanced web based tools and information technologies into existing administrative workflows to efficiently manage human capital and create strategic value for stakeholders (IBM, 2024). By digitizing core functions, modern organizations are actively transitioning their personnel departments from reactive support units into proactive drivers of business agility and overall performance.

Within large corporate infrastructures, this digital transformation acts as a catalyst for shifting from standardized administration to a dynamic talent architecture. Leveraging the Dynamic Capabilities Theory, academic research demonstrates that digital systems provide corporations with the strategic agility necessary to sense environmental changes, seize market opportunities, and reconfigure internal resources rapidly (Emerald Publishing, 2023). High performance work systems and high involvement work systems are increasingly supported by digital infrastructures that foster decentralized decision making and cross functional collaboration. Furthermore, this digitalization replaces intuitive management with concrete workforce analytics. By centralizing operations into comprehensive information systems, corporate leaders can utilize real time data to optimize workforce planning, analyze turnover trends, and track global productivity without relying on isolated spreadsheets (McKinsey, 2025). This transition also supports the modern physical and digital workplace reality, abandoning outdated annual performance reviews in favor of continuous listening systems and integrated learning platforms that enable self-directed professional development (Deloitte, 2025).

Conversely, for small and medium enterprises, embracing digital transformation is fundamentally an existential necessity to survive the crippling inefficiencies associated with manual administration. In these smaller environments, manual processes act as a severe growth blocker. Relying on physical paperwork creates a silent drain on organizational resources, with industry studies revealing that manual administrative waste costs organizations between four hundred and six hundred dollars per employee annually. According to Ernst and Young (2025), the financial burden of a single manual data entry task averages nearly five dollars, while complex workflows like processing payroll manually escalate these financial leakages

significantly. Beyond direct costs, smaller enterprises face massive security vulnerabilities and compliance nightmares when handling physical records. Strict regulatory frameworks currently mandate precise data portability and the explicit right to erasure (General Data Protection Regulation, 2016). Digital systems mitigate these risks by providing encrypted cloud storage and automated data retention policies, making strict regulatory compliance scalable. As these organizations grow, implementing digital employee portals empowers workers with self service capabilities, which significantly reduces administrative bottlenecks and fosters a more professional onboarding experience (InStaff, 2024).

However, the successful implementation of these digital platforms extends far beyond solving technological challenges. It is deeply intertwined with organizational behavior and employee psychology. Resistance to new information systems frequently emerges from a lack of technical confidence, a deep seated comfort with legacy routines, or inherent fears regarding job security as automation increases (EmployeeConnect, 2024). If an organization fails to clearly communicate the strategic benefits of the new system, this resistance can severely derail the entire implementation process. Drawing upon the Ability Motivation Opportunity framework and Social Cognitive Theory, researchers suggest that when employees perceive digital changes positively, they are significantly more likely to exhibit proactive change behaviors. This relationship is strongly mediated by work engagement, as engaged employees are more willing to invest the cognitive energy required to adapt to new digital workflows (ResearchGate, 2024). Interestingly, recent academic literature highlights that personal organization fit negatively moderates this relationship during digital transitions (PMC, 2024). When the cultural fit is exceptionally high, established routines feel highly comfortable, thereby diminishing the perceived urgency to enact change. Conversely, when the fit is lower, highly engaged employees are motivated to utilize new digital tools to proactively bridge operational gaps. Ultimately, to overcome these adoption hurdles, organizations must deploy inclusive change management interventions, ensuring that the workforce is actively engaged and supported throughout the transition.

1.1. Overview of Human Resource Management (HRM)

Human Resource Management (HRM) serves as the fundamental structural backbone that sustains the operational integrity and long term evolution of any contemporary enterprise. Within the increasingly complex paradigms of the global economy, the practice of HRM has successfully transcended its traditional administrative boundaries. Historically confined to routine functions such as payroll processing, record maintenance, and regulatory compliance, HRM has now emerged as a core strategic competency that fundamentally shapes organizational performance

and sustainable competitive advantage (Sari, Nisva, & Maulana, 2025). The most significant transformation in this field is the conceptual shift of the HR department from a conventional cost center to a proactive value driver (Sari et al., 2025). By meticulously aligning human capital practices with overarching corporate objectives, HR practitioners cultivate dynamic capabilities that empower organizations to reconfigure their internal resources and adapt with agility to market volatilities (Bamber, Bartram, & Stanton, 2017).

The success of any operational framework remains inherently dependent on the quality of its personnel. Consequently, the primary mandate of HRM is to ensure that the organization consistently maintains the appropriate talent at the correct positions with the specific skill sets required for optimal performance (Alam, Masum, Beh, & Hong, 2016). Amidst the challenges of global skill shortages and rising turnover rates, the strategic role of HRM becomes vital in designing talent acquisition frameworks and bridging competency gaps. Through the implementation of structured training programs, transparent career progression paths, and employee empowerment, HRM addresses the root causes of labor attrition while simultaneously elevating the overall quality of the workforce (Ravichandran & Dahiya, 2025).

Furthermore, HRM significantly dictates the employee experience through the application of High Performance Work Systems (HPWS) and High Involvement Work Systems (HIWS). These frameworks establish a sequence of professional practices that encourage active participation in decision making processes, thereby strengthening organizational commitment and engagement (Bamber et al., 2017). Such engagement is not merely a morale booster but a direct catalyst for superior business outcomes. Empirical research indicates that when HRM provides robust operational support mechanisms alongside effective work life balance policies, the organization achieves heightened sustainability and enhanced psychological resilience against professional pressures (Kanwar, 2025).

Moreover, the human element remains the epicenter of creativity and organizational innovation. Innovation is not an accidental occurrence but rather the product of a supportive environment facilitated by HRM acting as a catalyst for an inclusive culture and psychological safety. By embracing diversity regarding gender, age, and cultural backgrounds, HRM introduces fresh perspectives and innovative solutions to address complex challenges (Bamber et al., 2017). Modern management has also transitioned from rigid annual performance appraisals to a model of continuous feedback, coaching, and recognition, thereby fostering a perpetual learning ecosystem (Sari et al., 2025).

Finally, despite this strategic shift, the foundational role of HRM in risk management and legal compliance remains critical. Effective HR governance establishes transparent regulatory systems that prevent labor disputes and protect the corporate reputation from significant legal and financial repercussions. In summary, the true significance of HRM lies in its capacity to transform human capital into the most valuable asset of the firm. By fostering engagement, developing competencies, and building a culture of innovation, HRM creates a resilient foundation that allows the organization to survive, adapt, and achieve breakthroughs within a volatile business environment.

1.2. Overview of ERP Systems and the HR Module

1.2.1. Concepts of Enterprise Resource Planning (ERP)

Enterprise Resource Planning (ERP) is defined as a comprehensive management information system designed to facilitate the coordination and automation of core business functions within a unified platform and a single, centralized database. This holistic approach encompasses various domains, such as finance, human resources, manufacturing, and supply chain management. According to Davenport (1998), ERP serves as a cohesive software solution that integrates all departments and organizational functions, effectively acting as a critical digital infrastructure for the real-time synchronization of both internal and external corporate processes (Polivka & Dvorakova, 2021).

By eliminating data silos across departments, ERP systems optimize workflows and foster seamless cross-functional collaboration throughout the entire organization. This integration provides leadership with a holistic, real-time perspective alongside high-fidelity reporting, which facilitates data-driven strategic decision-making and enhances organizational agility in response to market volatility (NetSuite, n.d.). Furthermore, ERP systems play a vital role in process standardization and the optimization of scarce resources, including raw materials and labor. By minimizing operational errors and overhead costs, these systems ultimately bolster labor productivity and strengthen a firm's overall competitive advantage within the global marketplace.

1.2.2. Definition of Human Resource Information Systems (HRIS)

A Human Resource Information System (HRIS) is conceptualized as a systematic process or an integrated online software solution designed to facilitate the collection, input, storage, maintenance, retrieval, and validation of essential data regarding human resources and organizational personnel activities (Shaikh, 2014; Sharma, 2013 (Sharma, 2013). In terms of formal structure, an HRIS does not

exclusively denote a complex computerized framework. It may encompass informal mechanisms such as payroll records and timekeeping cards within small-scale enterprises, or conversely, expand into sophisticated digital personnel databases within large corporations (Sharma, 2013).

Within the landscape of contemporary management, an HRIS is regarded as a socio-technical program and a pivotal mechanism that automates workflows, enhances operational efficiency, and fosters data-driven decision-making (Albouti & Balaji, 2025) (Fiaz & Qureshi, 2024). Through its capacity for automated information integration and management, this system provides a robust platform that assists administrators in executing specialized HR functions and formulating strategic decisions pertaining to recruitment, placement, professional development, and resource allocation (Fiaz & Qureshi, 2024) ; Shaikh, 2014).

1.2.3. HRM within the ERP Environment

Within Enterprise Resource Planning (ERP) systems, the Human Resources (HR) module constitutes a pivotal element, systematically managing, automating, and refining processes throughout the employee lifecycle (Schwarz, 2025). Eschewing fragmented paper-based records or isolated Excel spreadsheets, this module operates as an integrated central repository, facilitating seamless synchronization of personnel data across functional domains including finance, accounting, and Customer Relationship Management (CRM) (Singh et al., n.d.; Schwarz, 2025). Such unification equips executive leadership with a comprehensive vantage point, illuminating the ramifications of labor expenditures and workforce efficacy on the organization's fiscal trajectory (Schwarz, 2025).

On the operational front, ERP-embedded HR management subsumes essential functionalities: oversight of employee data (encompassing personal dossiers, competency evaluations, and career advancement trajectories), monitoring of time and attendance (shift allocation and leave entitlements), automated payroll processing, and welfare program governance (Khan et al., 2025; Schwarz, 2025). Furthermore, the assimilation of electronic Human Resource Management (e-HRM) augments HR's purview to encompass e-recruitment, e-learning platforms, and online performance appraisal mechanisms, thereby instituting procedural uniformity and stringent adherence to extant regulatory frameworks (Almashyakhi, 2022).

Strategically, HR's integration into ERP delivers profound advantages by mechanizing routine administrative operations and markedly diminishing manual-induced discrepancies, thereby liberating HR practitioners for elevated pursuits such as strategic workforce forecasting, talent cultivation, and organizational

culture refinement (Schwarz, 2025). Concurrently, self-service portals for employees and sophisticated data analytics instruments not only enhance internal stakeholder satisfaction but also furnish decision-makers with empirical insights to optimize resource allocation and elevate productivity throughout the enterprise architecture (Almashyakhi, 2022; Schwarz, 2025).

1.2.3.1 Employee Management Module

By establishing a centralized database, the Employee Information Management module allows organizations to oversee the entire employee lifecycle within a unified framework. This digital consolidation of core personnel data, competency profiles, and performance metrics effectively mitigates the issue of information fragmentation (Schwarz, 2025). Furthermore, such a system grants leadership access to intuitive dashboards for monitoring critical human capital KPIs, including diversity ratios and revenue per employee (Schwarz, 2025). The inclusion of employee self-service (ESS) portals further empowers the workforce to manage their own data, a move that significantly bolsters organizational transparency and internal cohesion (IZISolution, n.d.; Schwarz, 2025).

1.2.3.2. Time and Attendance tracking

The time and attendance tracking mechanism functions as a pivotal module, providing a rigorous framework for measuring actual labor hours, which serves as a fundamental basis for resource allocation and cost containment (Schwarz, 2025). By digitalizing traditional timesheets, this sub-system automates the calculation of leave entitlements, absenteeism records, and overtime hours based on highly configurable parameters (IZISolution, n.d.; Schwarz, 2025). Furthermore, through real-time synchronization with biometric identification devices or mobile platforms, the system effectively mitigates administrative burdens, allowing HR professionals to pivot from manual data entry toward more strategic functions (IZISolution, n.d.).

1.2.3.3. Compensation and Benefits administration

Payroll automation represents the most critical output stage within an integrated human resource process. By leveraging automated data reconciliation algorithms that interface timekeeping systems with contractual obligations, this module facilitates precise compensation calculations, encompassing statutory tax withholdings and benefit contributions in strict adherence to prevailing regulatory standards (IZISolution, n.d.; Schwarz, 2025). Such automation not only mitigates the margin of human error but also ensures rigorous legal compliance, thereby minimizing fiscal risks associated with tax audits (Schwarz, 2025). Most notably, the instantaneous synchronization of personnel cost data with the broader ERP's financial

accounting module provides essential support for financial reporting and strategic budgetary planning (IZISolution, n.d.; Schwarz, 2025).

1.3. Data Visualization in Human Resource Management (HR Analytics)

1.3.1. Conceptual Framework and the Shift Toward Evidence Based Management

In the contemporary digital landscape, data visualization serves as a critical interface between complex analytical outputs and managerial cognition by transforming raw datasets into meaningful and actionable insights (Reddy & Pavithra, 2026). The transition from traditional human resource management practices toward HR Analytics signifies a fundamental evolution in utilizing evidence based practices to drive organizational strategy (Patil & Priya, 2025).

The psychological foundation of visualization lies in the human brain's ability to process visual information sixty thousand times faster than plain text, which enables decision makers to rapidly identify patterns, trends, and anomalies within vast repositories of human capital data (AlMobark, 2023). This capability is particularly vital as organizations migrate from descriptive analytics toward predictive and prescriptive models that anticipate future workforce needs rather than merely documenting historical events (Arellano et al., 2017).

1.3.2. The Structural Role of Business Intelligence (BI) in Data Integration

Business Intelligence represents a comprehensive ecosystem of technologies and processes designed to collect, store, and analyze data to optimize enterprise performance (Lạc Việt, 2026). Within the framework of modern human resource information systems, BI functions as the architectural layer that connects directly with the centralized databases of Enterprise Resource Planning (ERP) systems.

The most significant contribution of BI is its capacity to dismantle data silos by unifying information from disparate modules such as employee profiles, attendance records, and payroll systems. Through sophisticated platforms like Power BI or Tableau, raw data is refined and presented via interactive dashboards that allow users to perform multi level drill down analysis with high precision and real time updates (Dhivya & Vithyaramani, 2025). This integration ensures that HR reporting evolves from static documentation into a dynamic and consistent tool for supporting agile decision making across the entire organization.

1.3.3. HR KPIs and Strategic Value in Executive Decision Making

The core of effective HR data visualization resides in the establishment and monitoring of key performance indicators (KPIs) that measure both operational

efficiency and the return on investment (ROI) of human capital. Modern evaluation practices have moved beyond simple headcount tracking to focus on complex metrics such as attrition rates, cost per hire, labor productivity, and labor cost variance (Khaire & Alvekar, 2024).

Visualizing these KPIs facilitates a culture of evidence based management where leaders can move beyond intuition to validate their strategic choices. For instance, a BI dashboard can illuminate the direct correlation between compensation adjustments and department specific productivity shifts. This level of data transparency enhances the accuracy, speed, and confidence of executive decisions, thereby allowing the Human Resources function to operate as a genuine strategic business partner (Reddy & Pavithra, 2026). Ultimately, data visualization acts as the catalyst for converting complex workforce information into organizational intelligence and sustainable competitive advantage.

1.4. Theoretical Frameworks and Applied Methodologies

1.4.1. Business Process Reengineering in ERP Implementation

1.4.1.1. Definition of BPR

Business Process Reengineering (BPR) is theoretically grounded in the fundamental rethinking and radical redesign of organizational processes to achieve dramatic improvements in critical measures of performance. Unlike traditional management approaches that focus on incremental, piecemeal improvements, the core philosophy of BPR dictates a complete departure from existing, flawed procedures to achieve breakthrough enhancements. Pioneering scholars in the field emphasize the need to avoid merely automating obsolete manual tasks; instead, the methodology demands obliterating them and reconstructing work from the ground up. This radical transformation targets unprecedented improvements in both quantitative and qualitative criteria, such as productivity, failure tolerance, and operational flexibility. Consequently, BPR is not perceived as a series of marginal adjustments but as a strategic organizational imperative that fundamentally alters tasks and informational flows into highly integrated business processes

1.4.1.2. The Necessity of BPR during the Transition from Manual Management to ERP

The necessity for such radical redesign becomes particularly evident when an organization transitions from manual, paper-based, or disjointed spreadsheet management to a comprehensive Enterprise Resource Planning (ERP) system. Prior to ERP adoption, organizations often rely on isolated administrative systems or manual

tracking, which require continuous human intervention for data migration and are inherently prone to significant errors. Implementing an ERP system effectively necessitates reengineering existing business processes to align with the industry best practices that are fundamentally embedded within the software package. Rather than modifying the ERP software to accommodate the firm's current, potentially inefficient manual processes, BPR requires the organization to fundamentally redesign its structural operations to match the ERP package seamlessly. This structural realignment is crucial because translating fragmented manual tasks into a centralized, fully computerized platform establishes an integrated operational environment that prevents common human mistakes, eliminates redundant activities, and ensures data flows smoothly across functional boundaries

1.4.1.3. Performance Metrics Targeted by BPR in HRM: Processing Speed and Data Accuracy

Within the specific domain of Human Resources and Payroll administration, this synergistic application of BPR and ERP targets profound enhancements in distinct performance metrics. A primary objective of this reengineering effort is to drastically reduce operation process cycle times, thereby maximizing efficiency and accelerating critical functions such as payroll calculation and processing. Furthermore, because modern integrated systems rely heavily on a unified database where incorrect entries can trigger a negative domino effect across the entire enterprise, ensuring stringent data accuracy becomes a paramount performance measure. By obliterating paper-bound tracking and disjointed departmental communications, a reengineered HR process guarantees error-free records, ensures precise time management and overtime evaluation for payroll formulation, and provides instantaneous access to reliable personnel information. Ultimately, this fundamental shift from manual administration to a reengineered, technology-enabled framework allows organizations to realize quantum leaps in both payroll processing speed and overarching data integrity

1.4.2. Technology Acceptance Model (TAM)

1.4.2.1. Origins and Evolution of the TAM Model

Information systems research has long grappled with the fundamental paradigm of why individuals accept or reject new technological innovations within the workplace. To theoretically bridge the gap between technological potential and human behavioral implementation, Davis introduced the Technology Acceptance Model (TAM) in 1989 (Davis, 1989). The genealogical roots of this framework are firmly anchored in the Theory of Reasoned Action (TRA), a highly influential behavioral

psychology model conceptualized by Fishbein and Ajzen, which postulates that an individual's behavioral intention is the most immediate antecedent to their actual behavior (Fishbein & Ajzen, 1975). Recognizing that the broad parameters of the Theory of Reasoned Action required precise contextualization for information systems, Davis refined the model by focusing on specific, technology-centric cognitive beliefs that influence IT adoption behavior (Bazine, 2025). As enterprise technologies evolved into highly complex platforms, scholars recognized the necessity to expand the original framework. This imperative catalyzed the chronological evolution of the model, leading to the development of TAM2 by Venkatesh and Davis, which integrated crucial elements of social influence and cognitive instrumental processes (Venkatesh & Davis, 2000). Subsequently, Venkatesh and Bala promulgated TAM3, establishing a comprehensive architecture that incorporated the moderating effects of user experience and the degree of voluntariness in system usage (Venkatesh & Bala, 2008). The theoretical trajectory further matured into the Unified Theory of Acceptance and Use of Technology (UTAUT), which unified the insights of eight distinct adoption models to offer an exhaustive perspective on modern digital adoption across diverse technological environments (Venkatesh et al., 2003)

1.4.2.2. Core Components of the Model

At the theoretical core of the original Technology Acceptance Model lie two primary determinants that systematically dictate user attitude and subsequent behavioral intention: perceived usefulness and perceived ease of use (Davis, 1989). Perceived usefulness is academically conceptualized as the subjective probability or the degree to which an individual profoundly believes that utilizing a specific technological system will directly enhance their professional job performance (Davis, 1989). Within an organizational context, this cognitive construct captures the instrumental value of the technology, dictating the individual's definitive behavioral intention to use the system. Conversely, perceived ease of use delineates the extent to which a prospective user expects the operation of the target system to be entirely free of physical and mental exertion, effectively measuring the cognitive burden associated with technological navigation (Davis, 1989). The model mathematically and logically asserts a definitive causal linkage between these two constructs, demonstrating that perceived ease of use exerts a potent and direct influence on perceived usefulness (Bazine, 2025). When a software interface is intuitively designed and inherently straightforward to operate, the user expends considerably less cognitive bandwidth on deciphering system mechanics, which naturally and significantly elevates the system's perceived utility (Igbaria et al., 1995). In unison, these two fundamental cognitive beliefs forge the individual's subjective attitude toward the technology, which subsequently dictates their actual system usage and determines the overarching

success of the information system implementation (Davis, Bagozzi, & Warshaw, 1989)

1.4.2.3. External Variables Influencing ERP Acceptance

While perceived usefulness and perceived ease of use function as the central mediating mechanisms within the Technology Acceptance Model, their formation is heavily subordinated to a diverse array of external variables encompassing technological, individual, and organizational dimensions (Bazine, 2025). Comprehending these external precursors is an absolute necessity for contemporary enterprises deploying intricate infrastructures such as Enterprise Resource Planning (ERP) systems. On an individual cognitive level, the construct of computer self-efficacy plays an extraordinarily vital role, acting as a psychological buffer that actively diminishes technology-induced anxiety and directly elevates both perceived ease of use and perceived usefulness (Igbaria & Iivari, 1995). Furthermore, macroscopic organizational variables, predominantly characterized by the dual pillars of top management support and extensive training programs, formulate the environmental scaffolding required for successful ERP assimilation (Qachar & Nacer, 2023). Top management support furnishes the necessary strategic vision and confers structural legitimacy upon the project, motivating employees to embrace the technological shift and significantly influencing the implementation success (Umble, Haft, & Umble, 2003). Concurrently, comprehensive end-user training systematically dismantles operational ambiguities, transferring essential technical knowledge that demystifies the software architecture and substantially bolsters the perceived ease of use across the corporate hierarchy (Qachar & Nacer, 2023)

1.4.2.4. The Relationship between TAM and Digital Transformation in HRM

The profound theoretical tenets of the Technology Acceptance Model find their most critical practical application in the multifaceted arena of digital transformation within Human Resources and Payroll departments. The formidable transition from traditional, manual administrative architectures to a sophisticated, centralized ERP environment permanently disrupts established operational paradigms (Chertouras & Chatzikallia, 2004). Executing such a monumental transition necessitates rigorous Business Process Reengineering (BPR), which forcefully mandates the radical redesign and structural realignment of existing workflows to seamlessly integrate with the standardized best practices embedded within the global ERP package (Sumner, 1999). However, this systemic redesign frequently provokes intense organizational resistance, as employees accustomed to legacy methodologies may perceive the new algorithmic system as an exceedingly complex operational disruption or a direct existential threat to their job security (Al-Mashari & Zairi, 1999). The Technology

Acceptance Model elegantly elucidates that the sheer technological superiority of a reengineered process cannot unilaterally guarantee operational success; if the newly deployed interfaces are perceived by personnel administrators as excessively complicated, their behavioral intention to embrace the system will precipitously decline (Nikam & Prasad, 2017). Therefore, the successful culmination of a digital transformation demands a symbiotic, dialectical alignment between the structural rigor of business process reengineering on the system side and the cognitive perceptions modeled by TAM on the human side (Qachar & Nacer, 2023). By ensuring that the reengineered workflows are undeniably perceived as highly useful and demonstrably easy to navigate, organizational leaders can effectively cultivate positive technological acceptance and maximize the overarching efficacy of the enterprise (Umble, Haft, & Umble, 2003)

1.4.3. Integrated Theoretical Framework: The Intersection of Process Optimization (BPR) and Technology Acceptance (TAM)

1.4.3.1. Synergies between Process Structure and User Behavior

The implementation of comprehensive enterprise systems represents a monumental organizational paradigm shift that irrevocably alters corporate architecture. Within this transformative context, the structural configuration of business processes, representing the hard side of implementation, and the cognitive behavioral attitudes of the users, representing the soft side, must be conceptualized as two inextricably linked dimensions of the same endeavor (Al-Mashari & Zairi, 1999). Successful business process reengineering mandates a holistic equilibrium between technological infrastructure variables and human change management factors (Al-Mashari & Zairi, 1999). The technological architecture, embedded with global best practices, dictates the systemic efficiency, informational flow, and algorithmic logic of the enterprise. However, the actual realization of these theoretical structural benefits is entirely subordinate to the willingness of the human workforce to adopt and navigate the new operational parameters.

Scholarly literature corroborates this interdependence by demonstrating that enterprise implementations frequently falter not due to software deficiencies, but because of intense organizational resistance and a lack of cultural readiness (Qachar & Nacer, 2023). Radical process redesign intrinsically provokes a disruption of established comfort zones, triggering employee anxieties regarding job security, loss of traditional operational control, and fear of the unknown (Al-Mashari & Zairi, 1999; Qachar & Nacer, 2023). Consequently, an elegantly reengineered process remains functionally sterile and practically useless if the end-users actively reject the interface or circumvent the standardized workflows. Conversely, highly motivated and

technologically receptive users cannot achieve breakthrough enhancements in productivity if the underlying business processes remain structurally flawed, disjointed, or burdened by obsolete manual approvals. Therefore, achieving sustainable digital transformation demands a symbiotic alignment where optimized, reengineered systemic structures are perfectly matched by a workforce that is psychologically equipped and behaviorally motivated to harness the full capabilities of the new digital environment (Nikam & Prasad, 2017).

1.4.3.2. BPR as an Antecedent Impacting Perceived Usefulness (PU)

Moving deeper into the cognitive mechanics of user adoption, Business Process Reengineering serves as a critical structural antecedent to the core psychological constructs defined by the Technology Acceptance Model. When an organization transitions from fragmented, manual administrative tasks to a unified Enterprise Resource Planning environment, radical process redesign becomes an absolute necessity to align existing workflows with the standard business processes built into the software package (Qachar & Nacer, 2023). This structural realignment obliterates obsolete, redundant activities, centralizes decentralized data repositories, and fundamentally alters the execution of daily operations to prevent human error and accelerate processing velocity (Chertouras & Chatzikallia, 2004). In a meticulously reengineered environment, the newly streamlined workflows inherently reduce operational cycle times and eliminate cross-departmental informational friction, ensuring that complex tasks are executed with unprecedented precision (Chertouras & Chatzikallia, 2004).

From the psychological perspective of the end-user, this radical structural optimization translates directly into an elevated perception of the system's value. As originally postulated by Davis (1989), perceived usefulness is the subjective probability that a user believes employing a specific system will directly enhance their professional job performance (Bazine, 2025). When business process reengineering successfully eradicates tedious manual calculations, automates complex data aggregations, and provides instantaneous access to reliable information, the employee naturally and inevitably perceives the new technological system as a highly utilitarian instrument. The simplified and accelerated process structure serves as empirical proof to the user that the technology genuinely augments their workplace efficacy. Conversely, inadequate process redesign that merely automates flawed legacy operations will inevitably lead to systemic bottlenecks and user frustration, severely plunging the perceived utility of the system and precipitating widespread behavioral rejection. Thus, rigorous process optimization is not merely a technical prerequisite; it is the fundamental mechanism through which organizations tangibly demonstrate

system usefulness to their workforce, thereby directly influencing their behavioral intention to adopt the technology (Nikam & Prasad, 2017).

1.4.3.3. The Role of Training and Change Management in Bridging the Implementation Gap

Despite the theoretical harmony between optimized organizational processes and enhanced system utility, the practical transition from legacy methodologies to a reengineered digital landscape invariably produces a profound implementation gap. This gap is characterized by employee hesitation, technological anxiety, and the intrinsic human resistance to radical structural disruption (Al-Mashari & Zairi, 1999). To successfully bridge this critical divide between a newly redesigned system and a hesitant workforce, comprehensive change management and robust educational interventions emerge as indispensable catalysts (Qachar & Nacer, 2023). Effective change management strategies, anchored by visible top management support and extensive enterprise-wide communication, cultivate a receptive organizational culture that aligns individual expectations with the overarching corporate vision (Al-Mashari & Zairi, 1999). By transparently communicating the strategic imperatives and the specific benefits of the reengineered processes, leadership effectively mitigates the psychological uncertainties that fuel organizational resistance, fostering a shared belief in the system's value (Nikam & Prasad, 2017).

Concurrently, rigorous educational initiatives and hands-on training programs systematically dismantle technical ambiguities and foster crucial computer self-efficacy among the employees (Qachar & Nacer, 2023). Training interventions are not merely operational necessities; they are powerful psychological tools that directly shape user perception. Extensive and targeted training programs directly and profoundly elevate the users' perceived ease of use by equipping them with the cognitive frameworks and technical proficiencies necessary to navigate the new digital architecture with absolute confidence (Nikam & Prasad, 2017). Furthermore, reserving a substantial portion of the implementation budget specifically for end-user training significantly maximizes the probability of overarching project success, as it ensures that employees are fully capable of manipulating the sophisticated ERP interface without cognitive overload (Qachar & Nacer, 2023). Together, strategic change management and immersive training interventions dissolve psychological barriers and eliminate cognitive friction. They act as the vital connective tissue that transforms a skeptical, alienated workforce into proficient, willing participants, ultimately securing the synergistic triumph of both systemic reengineering and human technological acceptance.

CHƯƠNG 2: PHÂN TÍCH THỰC TRẠNG QUY TRÌNH NHÂN SỰ VÀ TIỀN LƯƠNG TẠI CÔNG TY TNHH HPA DIRECT VN

2.1. Giới thiệu tổng quan về Công ty TNHH HPA Direct VN:

- Cơ cấu tổ chức
- Quy mô nhân sự
- Mô hình hoạt động

2.2.1. Phân tích luồng nghiệp vụ Quản lý Hồ sơ & Hợp đồng.

2.2.2. Phân tích luồng nghiệp vụ Chấm công.

2.2.3. Phân tích luồng nghiệp vụ Tính lương & Chi trả.

2.2.4. Đánh giá công tác lập báo cáo và thống kê nhân sự hiện tại (Chỉ ra việc làm báo cáo Excel thủ công tốn thời gian và dễ sai sót).

2.3. Phân tích số liệu và Đánh giá hạn chế (Pain points):

- Đưa ra kết quả từ phỏng vấn và khảo sát để định lượng hóa các "nút thắt" hiện tại.

CHƯƠNG 3: ĐỀ XUẤT GIẢI PHÁP VÀ MÔ HÌNH HÓA HỆ THỐNG ERP

3.1. Khảo sát hệ thống

3.1.1. Mô tả bài toán

The problem description outlines the functional requirements for the proposed Enterprise Resource Planning system at HPA DIRECT VN COMPANY LIMITED. Upon joining the organization, every employee requires a formal labor contract and a comprehensive digital profile. The proposed system must empower Human Resource Executives to seamlessly create, store, and manage these contracts alongside the overall organizational structure. Furthermore, the human resources department is responsible for continuously updating employee records, tracking employment history, and managing position transfers or disciplinary actions throughout the lifecycle of each staff member within the company.

Concurrently, the daily operational flow necessitates a robust time and attendance mechanism. Employees must be able to digitally record their daily work hours, while human resource personnel manage work shifts, assign schedules, and

approve various leave requests. This centralized attendance data must then flow directly into the financial module without manual intervention. The system is required to automatically calculate monthly salaries by applying specific configuration parameters for basic wages, allowances, overtime pay, and mandatory tax deductions, thereby resolving the manual calculation bottlenecks previously handled by the accounting department.

To ensure holistic organizational governance, the proposed solution must feature advanced reporting and analytical capabilities. The architecture needs to provide the Board of Directors and human resource managers with automated dashboards that visualize real time data, enabling them to effortlessly search employee information and analyze payroll trends for strategic decision making. Finally, the system must include a secure self service portal, allowing individual employees to log into their personal accounts using their unique credentials to independently view their profiles, submit timesheet adjustments, and access their detailed payslips in a highly transparent manner.

3.1. Thiết kế quy trình nghiệp vụ mục tiêu (TO-BE Business Process):

- Mô hình BPMN cho luồng Hồ sơ & Hợp đồng mới.
- Mô hình BPMN cho luồng Chấm công tự động.
- Mô hình BPMN cho luồng Tính lương tích hợp.

3.2. Đặc tả Yêu cầu hệ thống (System Requirements):

3.2.1. Xác định và phân tích các chức năng chính của hệ thống

3.2.1.1. Chức năng quản lý hồ sơ nhân viên

This function serves as the centralized digital repository for all personnel records within the organization. It enables authorized human resource executives to systematically create, view, and archive comprehensive profiles for every staff member. By utilizing this feature, administrators can effectively oversee the entire workforce lifecycle, from the initial onboarding phase of new hires to the final offboarding processes, ensuring that all fundamental demographic and professional data is securely stored and easily accessible for official administrative purposes.

3.2.1.2. Cập nhật phòng ban

This system capability is designed to allow management to establish and continuously modify the internal organizational structure of the enterprise. It provides the necessary tools to create new departmental divisions, rename existing units, or

dissolve inactive teams to accurately reflect the current business hierarchy. Maintaining a precise department structure through this function is essential for defining correct reporting lines, facilitating internal communication workflows, and grouping employees accurately for subsequent performance and payroll evaluations.

3.2.1.3. Cập nhật thông tin nhân viên

To maintain the integrity and accuracy of the central personnel database, this function allows authorized users to modify specific details within an existing employee profile. Whenever a staff member experiences a change in personal circumstances, such as acquiring a new residential address, updating contact information, or changing bank account details for salary disbursement, human resource personnel can promptly input these modifications. This mechanism ensures that the enterprise always operates with the most current and verified employee data.

3.2.1.4. Quản lý hợp đồng lao động

This critical feature provides a structured digital framework for tracking and managing the legal employment agreements between the enterprise and its workforce. The system allows administrators to record specific contract parameters, including the official commencement date, expiration date, and the specific category of the employment agreement. Furthermore, the function is programmed to proactively alert human resource managers when a contract is approaching its expiration, thereby facilitating timely renewals, legal addendums, or formal terminations to ensure strict regulatory compliance.

3.2.1.5. Quản lý quá trình công tác

This function systematically records the professional trajectory of each employee throughout their tenure at the company. It serves as an official digital ledger for logging significant career milestones, including internal departmental transfers, official promotions, and periodic salary adjustments. By maintaining a chronological record of these operational events, the management team can accurately track employee development, evaluate long term performance, and ensure that all career progressions are transparently documented within the system.

3.2.1.6 Quản lý tính lương cho nhân viên

As a core financial component of the Enterprise Resource Planning architecture, this function automates the complex process of monthly salary computation. It seamlessly integrates verified attendance records, logged overtime hours, and approved leave days to calculate the accurate gross and net pay for each

staff member. The module automatically applies predefined configuration rules for tax deductions and mandatory insurance contributions, thereby significantly reducing the administrative burden on the accounting department and mitigating the financial risks associated with manual calculation errors.

3.2.1.7. Tìm kiếm

To enhance overall operational efficiency and system navigation, this utility provides a powerful data retrieval mechanism across the entire database. Authorized personnel can utilize this search capability to quickly locate specific personnel records or operational files by querying unique identifiers such as employee names, identification numbers, or contact details. This instant filtering capability eliminates the necessity for manual database scrolling, allowing administrators to access critical information promptly and accelerate daily administrative operations.

3.2.1.8. Thống kê báo cáo

This analytical function transforms raw operational data into actionable business intelligence for the executive leadership. It aggregates information from various system modules to generate comprehensive visual summaries regarding total workforce headcount, monthly payroll expenditures, and employee turnover metrics. Furthermore, the feature empowers management to export these detailed statistical reports into standardized digital formats, providing essential quantitative evidence to support strategic workforce planning and high level corporate decision making.

3.2.2. Đặc tả chức năng

3.2.3. Đặc tả phi chức năng

3.3. Phân tích hệ thống

3.3.1. Xác định tác nhân (Actors)

Actor	Use case	Description
GENERAL SYSTEM		
All Users	Login	Authenticate user identity using credentials (username and password) to grant access to authorized ERP modules.

HR RECORDS AND CONTRACTS		
HR Executive	Manage Organizational Structure	Initialize, update, and manage the company's hierarchy, including departments, job titles, and positions (Master Data).
	Manage Employee Profile	Create new accounts, update personal information, upload attachments (ID cards, degrees), and deactivate records for resigned employees.
	Manage Labor Contracts	Draft new contracts, track statuses, monitor expiration warnings, create contract addendums, and terminate contracts.
Employee	View Own Profile	Access the Employee Self-Service (ESS) portal to view personal information, current contracts, and employment history.
	Request Profile Update	Fill out a request form to modify personal data (address, bank account, tax code) and submit it to HR for approval.
MODULE 2: ATTENDANCE & TIMEKEEPER		
HR Executive	Manage Work Shifts	Define standard work shifts (e.g., morning, afternoon, administrative), set start/end times, and establish timekeeping rules.

	Assign Employee Schedules	Allocate fixed or rotating work shifts to specific departments or individual employees within the system.
	Monitor Attendance Records	View raw In/Out time data synchronized from timekeeping devices to identify late arrivals, early departures, and unexcused absences.
	Manage Leave Requests	Receive requests, check remaining leave balances, and approve or reject employee applications for leave, late arrivals, or early departures.
	Approve Timesheet Adjustments	Review and approve requests from employees explaining or asking to correct actual working hours (due to missed punches or device errors).
Employee	Clock In / Clock Out	Record the start and end times of daily work shifts via the system's mobile app or web portal.
	Submit Leave Request	Select the leave type (annual, sick, maternity), choose dates, and submit an absence request along with reasons or supporting documents.
	Request Timesheet Adjustment	Submit a request explaining timekeeping discrepancies and ask HR to correct the exact working hours for a specific day.

MODULE 3: PAYROLL		
HR Executive	Configure Payroll Parameters	Set up and update calculation formulas, including social/health insurance rates, Personal Income Tax (PIT) brackets, and allowance categories.
	Calculate Salary & Deductions	Trigger the system to automatically run the monthly payroll based on verified attendance data, automatically calculating gross/net pay, taxes, and insurance deductions.
Accountant	Approve Payroll Board	Comprehensively review the total payroll fund, audit anomalies, and lock the payroll board to forward to the Accounting department for disbursement.
	Distribute Payslips	Automatically generate and send detailed payslips to each employee's registered email or personal portal account.
Employee	View Payslip	Log in to review the breakdown of personal earnings (basic salary, allowances, overtime) and deductions (tax, social insurance) for a specific pay period.
MODULE 4: DASHBOARD & REPORTING		
Board of Directors	View HR Dashboard	Access real-time visual dashboards displaying human resource metrics, such as headcount,

		turnover rate, and leave balance fluctuations.
	Analyze Payroll & OT Trends	View charts analyzing and comparing the total payroll fund and overtime (OT) costs across different months or quarters to support financial control.
HR Executive	Export Statistical Reports	Extract detailed reports on employee directories, timekeeping data, and payroll boards into Excel/PDF formats for compliance and internal auditing.

3.2.2. Đặc tả Use Case và biểu đồ phân rã

3.3.2.1. Chức năng đăng nhập

Tên Usecase	Đăng nhập
Mã Usecase	UC-01
Tác nhân	Employee, HR Specialist, Accountant, Director
Mô tả	Actors log into the ERP system to access Human Resource Management and Payroll functions based on their assigned roles.
Trigger	Click on the "Login" button on the ERP system interface.
Precondition	The actor must have a valid account pre-created by the System Administrator.
Main Flow (Success)	

No.	Performed by	Action
1.	User	Select the Login function
2.	System	Display the Login interface.
3.	User	Enter Username/Email and

<table>
<tr><td></td><td>
<table>
<tr><td></td><td></td><td>Password (refer to data table *).</td></tr>
<tr><td>4.</td><td>User</td><td>Click the "Login" button.</td></tr>
<tr><td>5.</td><td>System</td><td>Validate that all mandatory fields are filled.</td></tr>
<tr><td>6.</td><td>System</td><td>Verify the credentials against the Database.</td></tr>
<tr><td>7.</td><td>System</td><td>Determine the User Role (Employee/HR/Accountant/Director).</td></tr>
<tr><td>8.</td><td>System</td><td>Redirect the user to the specific Dashboard according to their permissions.</td></tr>
</table>
</td></tr>
<tr><td>Alternative Flows</td><td>
<table>
<tr><td>No.</td><td>Performed by</td><td>Action</td></tr>
<tr><td>6a</td><td>System</td><td>Error message: "Please fill in all required fields" if any field is empty.</td></tr>
<tr><td>7a</td><td>System</td><td>Error message: "Invalid username or password" and prompt to retry.</td></tr>
<tr><td>7b</td><td>System</td><td>Error message: "Account is disabled" if the employee status is inactive.</td></tr>
</table>
</td></tr>
<tr><td>Post-condition</td><td>The actor successfully logs in and gains access to their authorized modules.</td></tr>
</table>

3.3.2.2. Manage Employee Profile

Use Case Name	Manage Employee Profile
Use Case ID	
Actors	HR Officer

Description	Allows the HR Officer to perform administrative tasks on employee records, including adding, viewing, updating, deactivating, and handling data imports/exports.
Trigger	HR Officer selects the "HR Records" module from the ERP dashboard.
Precondition	The HR Officer has successfully logged into the system with valid administrative credentials.
Main Flow (Success)	

No.	Performed by	Action
1.	HR Officer	Select "Manage Employee Profile" from the menu.
2.	System	Display the list of current employees and management options.
3.	HR Officer	Choose a specific sub-function: Add, View, Update, Deactivate, or Import/Export.
4.	System	[Include/Extend Logic]: System executes the specific sub-process (see sub-flows below).
5.	System	Validate changes and update the central HR database.
6.	System	Display a "Success" notification and refresh the employee list.

<table>
<tr><td>Sub-flows (Functional Extensions)</td><td>
<table>
<tr><td>Ref</td><td>Sub-function</td><td>Description</td></tr>
<tr><td>UC002.1</td><td>Add New Employee</td><td>Create a new record with personal and contract details.</td></tr>
<tr><td>UC002.2</td><td>View List / Details</td><td>Search, filter, and view comprehensive employee information.</td></tr>
<tr><td>UC002.3</td><td>Update Profile</td><td>Modify existing data (Addresses, Job Titles, etc.).</td></tr>
<tr><td>UC002.4</td><td>Deactivate Record</td><td>Set status to "Inactive" for resigned/terminate d staff.</td></tr>
<tr><td>UC002.5</td><td>Import / Export</td><td>Bulk upload via Excel or download data for reporting.</td></tr>
</table>
</td></tr>
<tr><td>Alternative Flow</td><td>
<table>
<tr><td>No.</td><td>Performed by</td><td>Action</td></tr>
<tr><td>5a</td><td>System</td><td>Error: "Duplicate Employee ID" if adding an existing ID.</td></tr>
<tr><td>5b</td><td>System</td><td>Error: "Invalid File Format" if importing a non-standard file.</td></tr>
</table>
</td></tr>
<tr><td>Post-condition</td><td>The employee database is updated, and all changes are logged for auditing.</td></tr>
</table>

3.3.2.3. Manage Employee Contracts

Use Case name	
Use Case ID	
Actors	
Description	Allows the HR Officer to handle the lifecycle of employment contracts, including creating new contracts, viewing history, updating terms, and exporting data.
Trigger	HR Officer selects the "Contract Management" module from the ERP sidebar.
Pre-condition	The HR Officer is authenticated and the specific employee profile already exists in the system.
Main Flow (Success)	

	HR Officer	Select "Manage Employee Contracts" for a specific employee.
	System	Display the contract overview (Active, Expired, and Pending contracts).
	HR Officer	Choose a sub-function: Add New, View, Update, or Export.
	System	[Include Logic]: If adding or updating, the system triggers "Manage Attachments" to upload the scanned contract.
	HR Officer	Enter or modify contract terms

<table>
<tr><td></td><td>
<table>
<tr><td></td><td></td><td>(Start date, End date, Salary grade).</td></tr>
<tr><td></td><td>System</td><td>Validate the contract dates (e.g., End date must be after Start date).</td></tr>
<tr><td></td><td>System</td><td>Save the contract details and update the employee's payroll status.</td></tr>
</table>
</td></tr>
<tr><td>Sub-flows (Functional Extensions)</td><td>
<table>
<tr><td></td><td></td><td></td></tr>
<tr><td>UC003.1</td><td>Add New Contract</td><td>Define a new legal agreement between HPA and the employee.</td></tr>
<tr><td>UC003.2</td><td>View List / Details</td><td>View historical contract versions and current terms.</td></tr>
<tr><td>UC003.3</td><td>Update Contract</td><td>Renew a contract or adjust terms via an Addendum.</td></tr>
<tr><td>UC003.4</td><td>Export Contract</td><td>Generate a PDF/Excel file of the contract for physical signing.</td></tr>
</table>
</td></tr>
<tr><td>Alternative Flows</td><td>
<table>
<tr><td></td><td></td><td></td></tr>
<tr><td>6a</td><td>System</td><td>Error: "Date Conflict" if the new contract overlaps with an</td></tr>
</table>
</td></tr>
</table>

	existing one.
Postcondition	The contract record is updated in the Database, and the Payroll module is notified of any salary changes.

3.3.2.3. Biểu đồ Use Case cho phân hệ Time & Attendance.

Use Case Name	Manage Employee Attendance
Use Case ID	UC-04
Actors	HR Officer
Description	Allows the HR Officer to oversee all timekeeping activities, including shift configuration, scheduling, leave management, and manual data corrections to ensure accurate payroll input
Trigger	HR Officer navigates to the "Attendance & Timekeeping" module in the ERP system.
Precondition	The HR Officer has administrative access to the HR Attendance module.
Main Flow (Success)	

No.	Performed by	Action
	HR Officer	Access the "Manage Employee Attendance" dashboard.
	System	Display real-time attendance statistics and management options.
	HR Officer	Select one of the extended sub-functions (Shifts,

		Schedules, Records, Leave, or Adjustment).
	System	Open the specific interface for the selected functional area.
	HR Officer	Perform necessary data entries or approvals.
	System	Validate the entries and update the central attendance database.
	System	Log the changes for audit purposes and notify relevant employees if necessary.

	Sub-function	Description
UC004.1	Manage Work Shifts	Define working hours, break times, and grace periods for different shifts.
UC004.2	Assign Schedules	Map employees or departments to specific work shifts and rotations.
UC004.3	Manage Attendance Records	Monitor daily check-in/out logs and track late

<table>
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<table>
<tr><td></td><td></td><td>arrivals/early departures.</td></tr>
<tr><td>UC004.4</td><td>Manage Leave Requests</td><td>Review, approve, or reject employee leave applications (Sick, Annual, Unpaid).</td></tr>
<tr><td>UC004.5</td><td>Manage Timesheet Adjustment</td><td>Manually correct logs due to technical errors or forgotten check-ins.</td></tr>
</table>
</td></tr>
<tr><td>Alternative Flows</td><td>
<table>
<tr><td>No.</td><td>Performed by</td><td>Action</td></tr>
<tr><td>6a</td><td>System</td><td>Error: "Schedule Conflict" if an employee is assigned to overlapping shifts.</td></tr>
<tr><td>6b</td><td>System</td><td>Error: "Insufficient Leave Balance" if a leave request exceeds available days.</td></tr>
</table>
</td></tr>
<tr><td>Postcondition</td><td>The timekeeping data is finalized and synchronized with the Payroll module for salary processing.</td></tr>
</table>

3.3.2.3. Biểu đồ Use Case cho phân hệ Payroll.

3.3.2.4. Biểu đồ Use Case cho phân hệ Dashboard & Reporting (Phân quyền riêng cho Ban Giám đốc và HR Manager).

3.4. Cấu trúc Dữ liệu & Quy tắc nghiệp vụ (Data & Business Rules):

- Ma trận phân quyền CRUD và công thức tính lương/thuế.

3.5. Thiết kế Hệ thống Báo cáo Quản trị (Management Dashboard):

3.5.1. Xác định các chỉ số đo lường trọng yếu

- Xây dựng bộ KPIs như Quỹ lương
- Tỷ lệ OT
- Tỷ lệ vắng mặt
- Cơ cấu nhân sự

3.5.2. Thiết kế giao diện trực quan hóa dữ liệu:

- Mockup hoặc báo cáo xuất từ Power BI/Tableau dựa trên dữ liệu mẫu.

KẾT LUẬN VÀ KIẾN NGHỊ

- Tóm tắt các kết quả đạt được.
- Những hạn chế của đề tài và hướng nghiên cứu trong tương lai.

PHỤ LỤC

- Bảng câu hỏi,
- Mẫu phỏng vấn
- Dummy data dùng cho Dashboard.

DANH MỤC TÀI LIỆU THAM KHẢO